

Research Experience Report

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Research Experience

I am currently working at the **Center for Quantum Science and Technology** at the University of Isfahan. My research and training are focused on the fabrication and characterization of integrated photonic devices based on lithium niobate (LiNbO_3) crystals.

As part of my work, I have gained hands-on experience in photonic device fabrication in the cleanroom. I have learned lithography techniques, including direct-write lithography for patterning crystals, as well as UV-based lithography using pre-designed masks to fabricate optical waveguides.

At present, my research is divided into two major experimental areas: **poling** and **dissipation**.

Poling Setup

In the poling experimental station, I am working on periodic poling of magnesium-doped lithium niobate ($\text{MgO}:\text{LiNbO}_3$) crystals. We investigate two distinct approaches:

1. **Metallic-electrode poling:** patterned electrodes are deposited on the crystal surface and a high-voltage electric field is applied.
2. **Liquid-electrolyte poling:** an electrolyte solution is used to establish contact with the crystal surface, enabling domain inversion under high-voltage bias.

By optimizing the parameters in both approaches, our goal is to achieve high-quality domain inversion suitable for efficient spontaneous parametric down-conversion (SPDC) and the generation of entangled photon pairs.

Dissipation Setup

In the dissipation experimental station, I am working on a free-space setup designed to measure optical losses of fabricated waveguides. The propagation loss is measured using a 1550 nm laser, while thermal effects are studied with a hot plate. This enables us to characterize the waveguide loss coefficient at different temperatures and better understand the thermal stability of the devices.

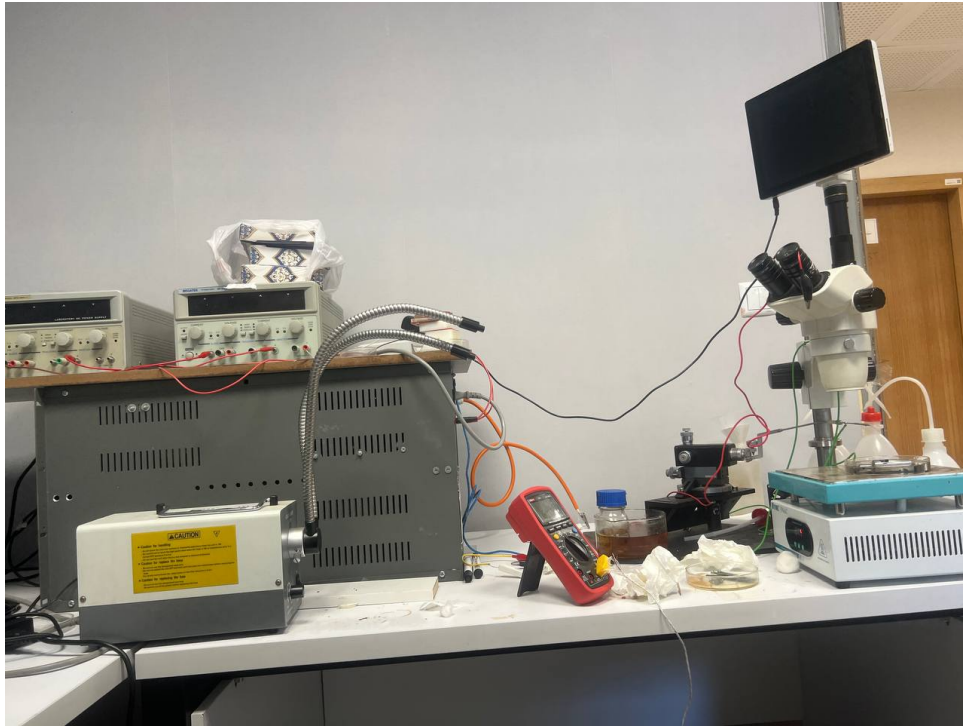


Figure 1: Experimental setup for poling $\text{MgO}:\text{LiNbO}_3$ crystals using metallic electrodes and electrolyte-based methods.

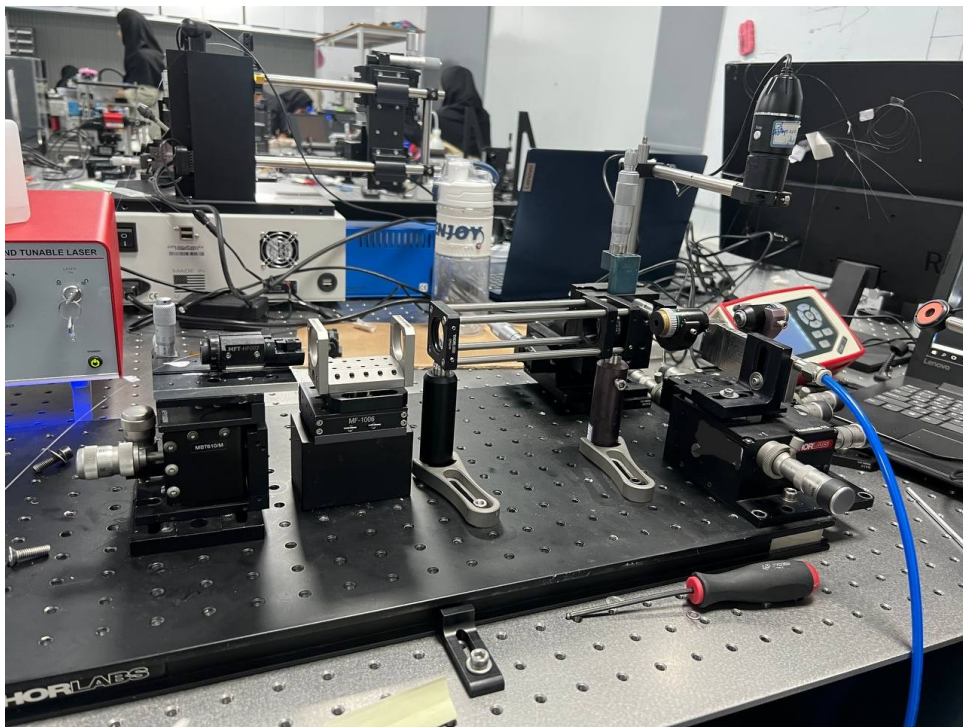


Figure 2: Free-space dissipation setup used for waveguide loss measurements at different temperatures.